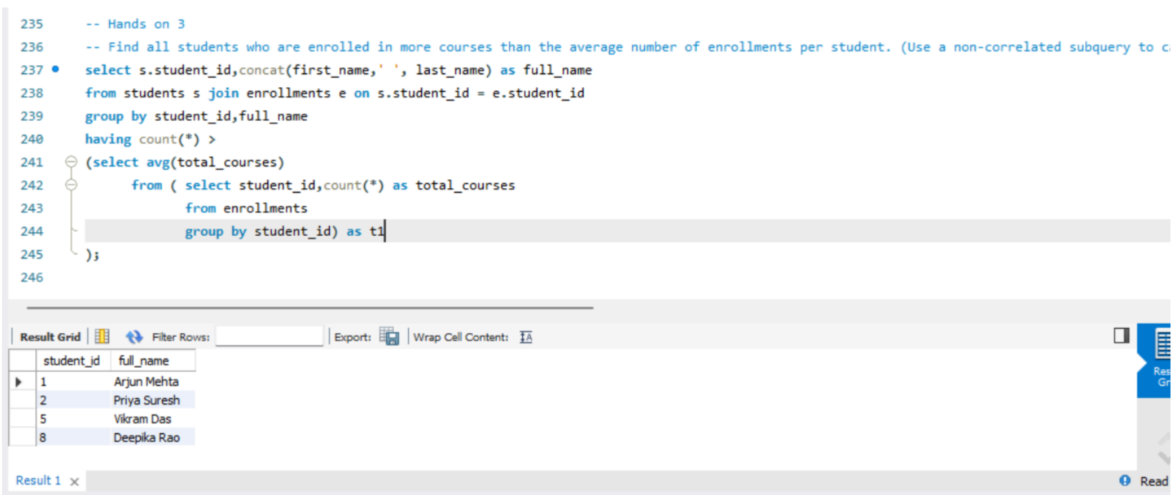
Hands on 3

Task 2

Task 1:

Find all students who are enrolled in more courses than the average number of enrollments per
student. (Use a non-correlated subquery to calculate the average.)



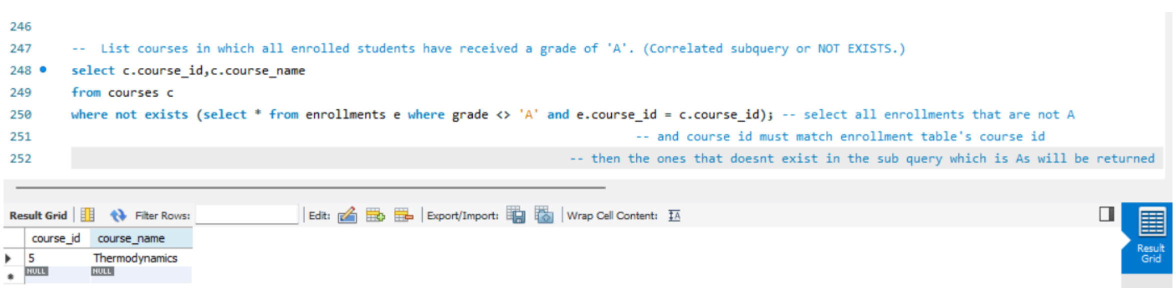
```sql
-- Hands on 3
-- Find all students who are enrolled in more courses than the average number of enrollments per student. (Use a non-correlated subquery to c
select s.student_id,concat(first_name,' ', last_name) as full_name
from students s join enrollments e on s.student_id = e.student_id
group by student_id,full_name
having count(*) >
(select avg(total_courses)
     from ( select student_id,count(*) as total_courses
            from enrollments
            group by student_id) as t1
);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id	full_name
1	Arjun Mehta
2	Priya Suresh
5	Vikram Das
8	Deepika Rao

Result 1 × Read

List courses in which all enrolled students have received a grade of 'A'. (Correlated subquery or NOT
EXISTS.)



```sql
--  List courses in which all enrolled students have received a grade of 'A'. (Correlated subquery or NOT EXISTS.)
select c.course_id,c.course_name
from courses c
where not exists (select * from enrollments e where grade <> 'A' and e.course_id = c.course_id); -- select all enrollments that are not A
                                                          -- and course id must match enrollment table's course id
                                                    -- then the ones that doesnt exist in the sub query which is As will be returned
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

course_id	course_name
5	Thermodynamics
NULL	NULL

Find the professor with the highest salary in each department using a correlated subquery

```sql
-- Find the professor with the highest salary in each department using a correlated subquery
select p1.professor_id,p1.prof_name,p1.department_id,p1.salary
from professors p1
where p1.salary = (select max(salary) from professors p2 where p1.department_id = p2.department_id);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

professor_id	prof_name	department_id	salary
1	Dr. Anand Krishnan	1	95000.00
3	Dr. Sunil Rajan	2	82000.00
4	Dr. Latha Gopal	3	79000.00
5	Dr. Kartik Bose	4	76000.00
NULL	NULL	NULL	NULL

Using a subquery in the FROM clause (derived table), calculate the per-department average salary
and then filter to departments where that average exceeds 85,000.

```sql
--  Using a subquery in the FROM clause (derived table), calculate the per-department average salary and then filter to departments where that averag
select *
from( select department_id,avg(salary) as avg_salary from professors group by department_id) as d_t
where avg_salary > 85000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_id	avg_salary
1	91500.000000

Result Grid